

# Probability Insights Report

Student Competitive Exam Dataset

|          |          |           |
|----------|----------|-----------|
| 200      | 5        | 71%       |
| Students | Features | Pass Rate |

## Topics Covered

- Basic Probability & Events
- Random Variable & Binomial Distribution
- Venn Diagram & Set Theory
- Contingency Table & Conditional Probability
- Bayes' Theorem Application
- Final Summary & Key Insights

## 1. Dataset Overview

The dataset contains **200 student records** with 5 features and a binary exam outcome (Pass / Fail).

| Feature             | Type        | Description                     |
|---------------------|-------------|---------------------------------|
| study_hours         | Numeric     | Weekly study hours (per week)   |
| attendance          | Numeric     | Class attendance percentage (%) |
| group_discussion    | Categorical | Participation: Yes / No         |
| previous_test_score | Numeric     | Score in previous test          |
| final_exam_pass     | Categorical | Binary outcome: Pass / Fail     |

### Dataset Split

| Category | Count | Percentage | Note           |
|----------|-------|------------|----------------|
| Pass     | 142   | 71%        | Majority class |
| Fail     | 58    | 29%        | Minority class |
| Total    | 200   | 100%       |                |

## 2. Basic Probability Events (Q1 & Q2)

### Formula

$$P(\text{Event}) = \text{Number of favourable outcomes} / \text{Total outcomes}$$
$$P(A) = n(A) / n(S)$$

### Three events from the dataset

| Event                     | Calculation | Result | %     |
|---------------------------|-------------|--------|-------|
| P(Pass)                   | 142 / 200   | 0.71   | 71%   |
| P(Group Discussion = Yes) | 99 / 200    | 0.495  | 49.5% |
| P(Attendance > 75%)       | 101 / 200   | 0.505  | 50.5% |

### Empirical vs. Theoretical Probability

Empirical:  $P(\text{Pass}) = 142/200 = 0.71$  (from actual data)

Theoretical:  $P(\text{Pass}) = 1/2 = 0.50$  (assuming equal likelihood)

Difference =  $0.71 - 0.50 = +0.21$  (Empirical > Theoretical)

★ Empirical probability (0.71) is 21 points above theoretical (0.50). Real students pass at a rate well above chance — preparation genuinely matters.

### 3. Random Variable & Binomial Distribution (Q3)

#### Setup

| Symbol       | Value               | Meaning                     |
|--------------|---------------------|-----------------------------|
| X            | # of Passes         | Random variable (discrete)  |
| n            | 3                   | Students selected per trial |
| p            | 0.71                | P(Pass) from dataset        |
| q = 1-p      | 0.29                | P(Fail)                     |
| Distribution | $X \sim B(3, 0.71)$ | Binomial with n=3, p=0.71   |

#### Binomial PMF Formula

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$$

Where:  $C(n, k) = n! / [k! \times (n-k)!]$  (Combination)

p = probability of success

k = number of successes

n = total trials

#### Step-by-Step Probability Distribution

| k (Passes) | Full Calculation                                                   | P(X = k) |
|------------|--------------------------------------------------------------------|----------|
| X = 0      | $C(3,0) \times 0.71^0 \times 0.29^3 = 1 \times 1 \times 0.024389$  | 0.0244   |
| X = 1      | $C(3,1) \times 0.71^1 \times 0.29^2 = 3 \times 0.71 \times 0.0841$ | 0.1791   |
| X = 2      | $C(3,2) \times 0.71^2 \times 0.29^1 = 3 \times 0.5041 \times 0.29$ | 0.4386   |
| X = 3      | $C(3,3) \times 0.71^3 \times 0.29^0 = 1 \times 0.3579 \times 1$    | 0.3579   |
| Total      | Sum of all probabilities                                           | 1.0000   |

#### Mean and Variance

Mean ( $\mu$ ) =  $n \times p = 3 \times 0.71 = 2.13$   
Variance ( $s^2$ ) =  $n \times p \times (1-p) = 3 \times 0.71 \times 0.29 = 0.6177$   
Std Dev ( $s$ ) =  $\sqrt{s^2} = \sqrt{0.6177} = 0.786$

★ On average 2.13 out of 3 students pass. The highest probability is at  $X=2$  ( $P=0.4386$ ), meaning most likely 2 out of 3 will pass.

## 4. Venn Diagram — Study Hours & Attendance (Q4)

### Set Definitions & Counting Formulas

$A = \{ \text{students: Study Hours} > 10/\text{week} \}$   
 $B = \{ \text{students: Attendance} > 80\% \}$   
  
 $n(A \text{ only}) = n(A) - n(A \cap B) = (92+60) - 60 = 92$   
 $n(B \text{ only}) = n(B) - n(A \cap B) = (60+18) - 60 = 18$   
 $n(A \cap B) = 60$  (both conditions met)  
 $n(A \cup B) = 92 + 60 + 18 = 170$   
 $n(\text{Neither}) = 200 - 170 = 30$   
Check:  $92 + 60 + 18 + 30 = 200$  VERIFIED

### Region Breakdown

| Region                     | Count | %    | Description                  |
|----------------------------|-------|------|------------------------------|
| A only (Study>10, Att≤80%) | 92    | 46%  | Hard workers, low attendance |
| A ∩ B (both conditions)    | 60    | 30%  | Most disciplined students    |
| B only (Att>80%, Study≤10) | 18    | 9%   | Regular attenders, low study |
| Neither                    | 30    | 15%  | Low study & low attendance   |
| Total                      | 200   | 100% | All students                 |

### Probabilities from Venn Regions

$P(A) = 152/200 = 0.76$   $P(B) = 78/200 = 0.39$   
 $P(A \cap B) = 60/200 = 0.30$   $P(A \cup B) = 170/200 = 0.85$   
 $P(\text{Neither}) = 30/200 = 0.15$

★ 60 students (30%) satisfy both conditions — the ideal student profile combining high study hours AND high attendance.

## 5. Contingency Table & Probability (Q5)

Contingency Table: Group Discussion vs. Exam Result

| Group Discussion | Fail | Pass | Row Total |
|------------------|------|------|-----------|
| No               | 35   | 66   | 101       |
| Yes              | 23   | 76   | 99        |
| Total            | 58   | 142  | 200       |

### Probability Formulas & Calculations

1. JOINT Probability  
 $P(A \text{ and } B) = n(A \text{ and } B) / n(S)$   
 $P(\text{GD=Yes AND Pass}) = 76 / 200 = 0.38$

2. MARGINAL Probability  
 $P(\text{Pass}) = 142 / 200 = 0.71$   
 $P(\text{GD=Yes}) = 99 / 200 = 0.495$

3. CONDITIONAL Probability  
 $P(A|B) = P(A \text{ and } B) / P(B) = n(A \text{ and } B) / n(B)$   
 $P(\text{Pass} \mid \text{GD=Yes}) = 76 / 99 = 0.7677 \text{ (76.77\%)}$   
 $P(\text{Pass} \mid \text{GD=No}) = 66 / 101 = 0.6535 \text{ (65.35\%)}$

### Independence Check

INDEPENDENT if:  $P(A \wedge B) = P(A) \times P(B)$

$P(\text{GD=Yes AND Pass}) = 0.38$   
 $P(\text{GD=Yes}) \times P(\text{Pass}) = 0.495 \times 0.71 = 0.3515$

$0.38 \neq 0.3515 \Rightarrow$  Events are DEPENDENT (not independent)  
Can co-occur  $\Rightarrow$  NOT Mutually Exclusive

★ Group discussion boosts pass probability from 71% (base) to 76.77% — an 11.42 percentage point improvement over non-participants.

## 6. Bayes' Theorem — Attendance & Passing (Q7)

### Formula

$$P(A|B) = [ P(B|A) \times P(A) ] / P(B)$$

Applied here:

$$P(\text{Pass}|H) = [ P(H|\text{Pass}) \times P(\text{Pass}) ] / P(H)$$

Where H = High Attendance event (> 80%)

Given Values

| Symbol    | Value  | Meaning                                 |
|-----------|--------|-----------------------------------------|
| P(H Pass) | 0.70   | 70% of passers had high attendance      |
| P(H Fail) | 0.40   | 40% of failers had high attendance      |
| P(H)      | 0.60   | 60% of all students had high attendance |
| P(Pass)   | 0.71   | Overall pass rate (142 / 200)           |
| P(Pass H) | 0.8283 | RESULT — 82.83% pass probability        |

Step-by-Step Calculation

Step 1: Numerator =  $P(H|\text{Pass}) \times P(\text{Pass})$   
=  $0.70 \times 0.71$   
= 0.497

Step 2: Denominator =  $P(H) = 0.60$   
Verification via Total Probability Rule:  
 $P(H) = P(H|\text{Pass}) \times P(\text{Pass}) + P(H|\text{Fail}) \times P(\text{Fail})$   
=  $0.70 \times 0.71 + 0.40 \times 0.29$   
=  $0.497 + 0.116 = 0.613 \sim 0.60$  (given)

Step 3:  $P(\text{Pass}|H) = 0.497 / 0.60$   
= 0.8283

RESULT:  $P(\text{Pass} \mid \text{High Attendance}) = 82.83\%$

★ 82.83% is the highest probability in this entire analysis. A student with >80% attendance is nearly certain to pass.

7. Final Summary — Key Findings

Ranked Factors Affecting Pass Probability

| Rank | Factor                     | Method                  | Pass % |
|------|----------------------------|-------------------------|--------|
| 1    | High Attendance (>80%)     | Bayes' Theorem          | 82.83% |
| 2    | Study>10h + Attendance>80% | Venn Diagram            | Strong |
| 3    | Group Discussion = Yes     | Conditional Probability | 76.77% |
| 4    | Overall Base Rate          | Empirical Probability   | 71.00% |
| 5    | Group Discussion = No      | Conditional Probability | 65.35% |

Key Takeaways

- Attendance is the STRONGEST predictor — Bayes' Theorem: students with >80% attendance have an 82.83% chance of passing.
- Group Discussion participation boosts pass rate from 71% to 76.77% — an 11.42 percentage point gain over non-participants.
- 60 students (30%) had BOTH high study hours AND high attendance — the ideal student profile (A intersection B).
- Empirical P(Pass) = 0.71 vs. Theoretical = 0.50 — the 21-point gap proves preparation has a real impact.
- Binomial mean = 2.13 confirms that in any random group of 3 students, passing is the most likely outcome.
- X=2 has the highest Binomial probability (0.4386) — most likely, 2 out of 3 randomly chosen students will pass.

Probability Concepts Applied

| Concept                   | Where Applied                                     |
|---------------------------|---------------------------------------------------|
| Empirical Probability     | Pass rate, Group Discussion rate, Attendance rate |
| Theoretical Probability   | Compared with empirical P(Pass)                   |
| Binomial Distribution     | P(X passes) from 3 random students, n=3, p=0.71   |
| Venn Diagram / Set Theory | Study Hours vs. Attendance overlap analysis       |
| Joint & Marginal Prob.    | Contingency table — Group Discussion vs. Pass     |
| Conditional Probability   | P(Pass GD=Yes) and P(Pass GD=No)                  |
| Independence Test         | Verified GD and Pass are dependent events         |
| Bayes' Theorem            | P(Pass   High Attendance) = 82.83%                |